

## Probability Theory

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Augusts 31th, 2026

Solve the following problem.

### Box 1: Application Example (Was it the burglar?)

Mary lives in San Francisco. One afternoon, she is driving back home and receives a phone call from her neighbor Jane. She told her that her house alarm was set off (A). While driving, she also heard on the radio (R) that a small earthquake (E) hit the city. Small earthquakes sometimes activate the alarm, and perhaps this is the reason why the alarm was sounding.

1. What is the probability that a burglar (B) broke into the house?
2. What is the probability that a burglar broke into the house given that the alarm was set off?
3. What is the probability that a burglar broke into the house given that the alarm was set off and a small earthquake hit the city?
4. Assume that there is a probability that Jane may not have listened to the alarm ( $J^c$ ) What is the probability that a burglar broke into the house given that Jane listened to the alarm?

To solve this problem define the random variables  $R(R) = 1$ ,  $R(R^c) = 0$ ,  $A(A) = 1$ ,  $A(A^c) = 0$ ,  $E(E) = 1$ ,  $E(E^c) = 0$ , and  $B(B) = 1$ ,  $B(B^c) = 0$ ,  $J(J) = 1$ , and  $J(J^c) = 0$ .